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UNNI C02 SMARTIFICATION – REV A

Two-sheet cleanup: standard power symbols, grid-aligned wiring, explicit local connectivity.

USB + Power

ESP32–C6 + Auxiliary

Simulation Harness

File: 01\_usb\_power.kicad\_sch

File: 02\_esp\_aux.kicad\_sch

File: 03\_simulation\_harness.kicad\_sch

Unni C02 Smartification

Sheet: /  
File: unni-smartification-c6.kicad\_sch

Title: Unni C02 Smartification – ESP32–C6 Rev A

Size: A3Date: 2026–08–19

KiCad E.D.A. 10.0.5Id: 1/4

A

B

C

D

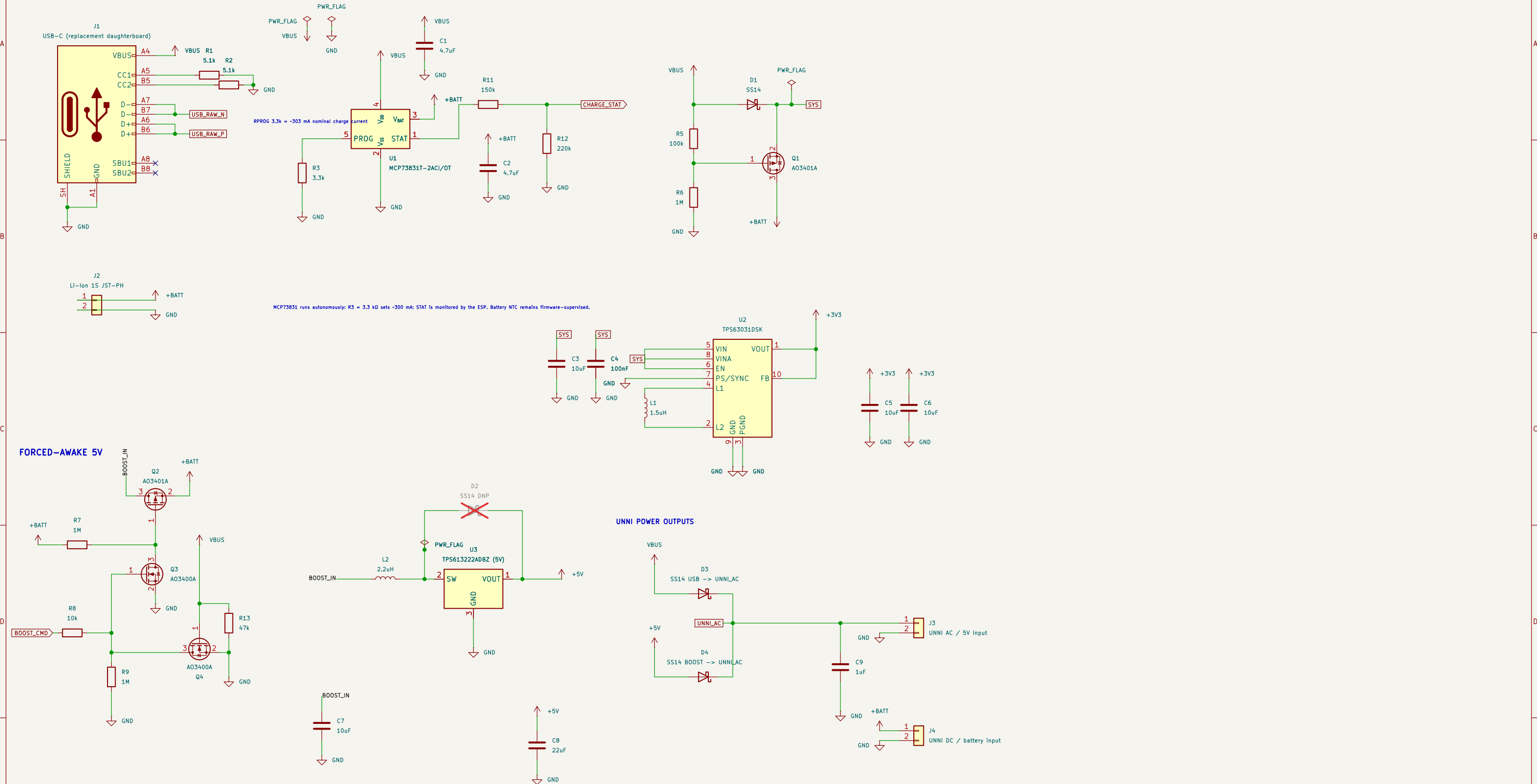
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|----------------------------------------------------|------------------|----------------------|
| Unni C02 Smartification                            |                  |                      |
| Sheet: /<br>File: unni-smartification-c6.kicad_sch |                  |                      |
| Title: Unni C02 Smartification – ESP32–C6 Rev A    |                  |                      |
| Size: A3                                           | Date: 2026–08–19 | Rev: A–v29–validated |
| KiCad E.D.A. 10.0.5                                | Id: 1/4          |                      |

USB + POWER

All placement/wiring is on 50 mil (1.27 mm) grid. Standard KiCad symbols used where available.



ESP32-C6 + AUXILIARY CIRCUITS

GPIO0-3 are reserved for ADC inputs; RT/RH use RTC-capable GPIO6/7 for deep-sleep wake.

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A

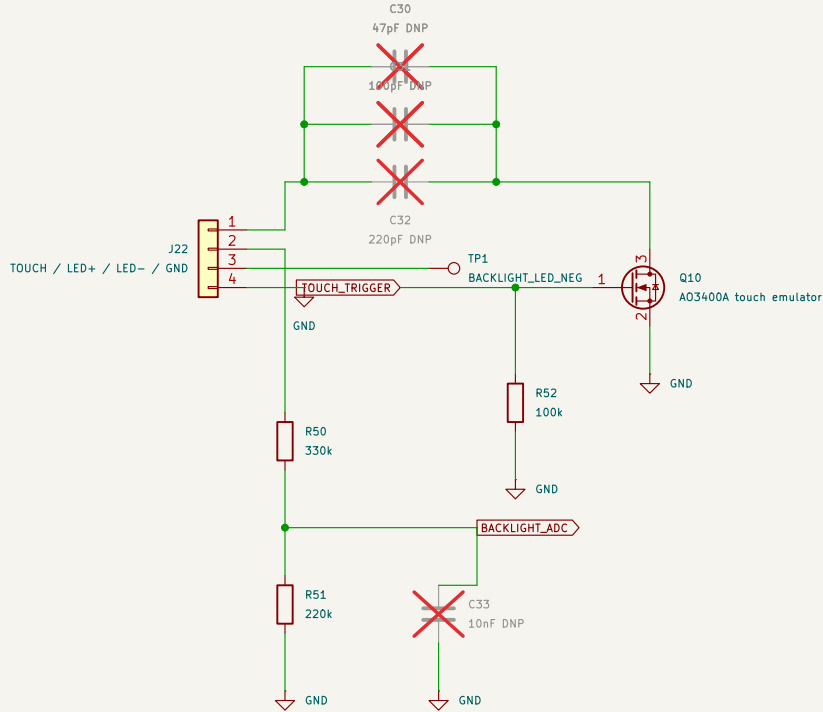
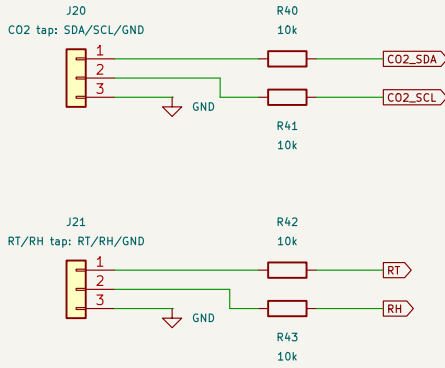
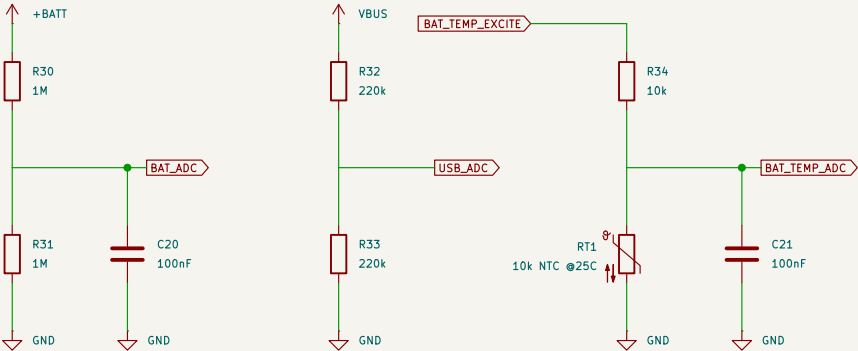
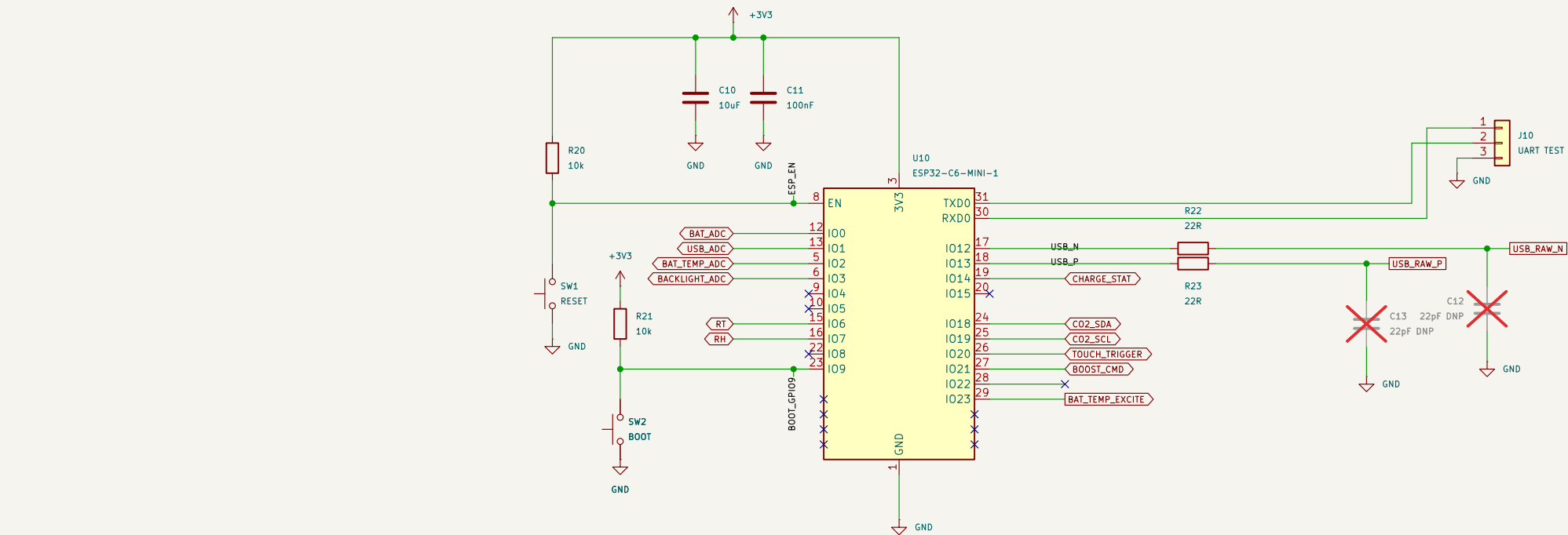
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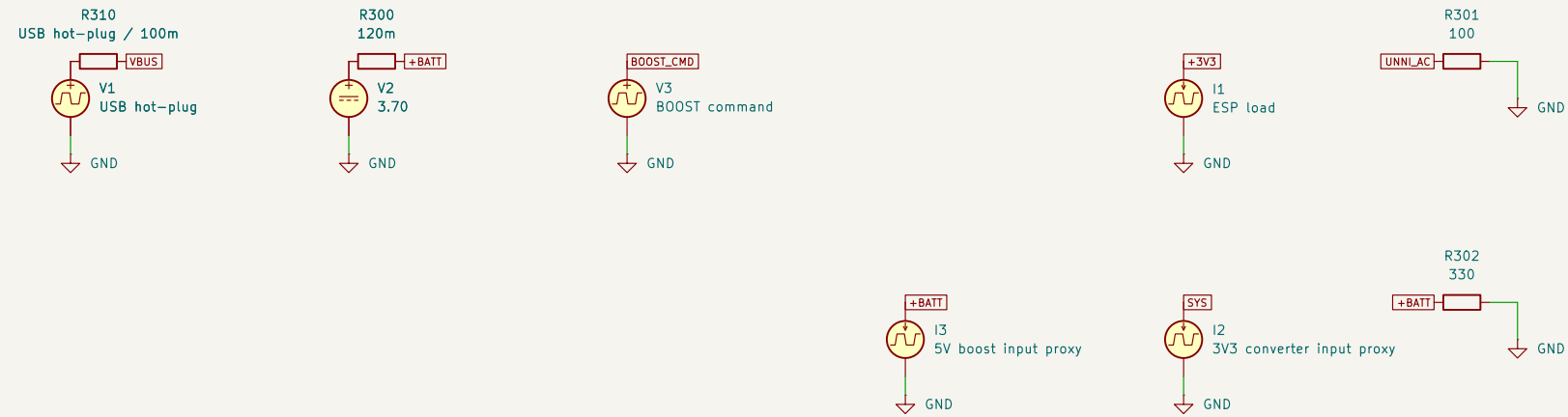
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**SIMULATION HARNESS – excluded from BOM/PCB**

Scenario: battery-only → forced awake → USB hot-plug while BOOST\_CMD remains asserted → autonomous charging → USB removal. Battery model includes 120 mΩ ESR. Battery model includes 120 mΩ ESR. I2 is the 3V3-converter input-power proxy. TPS613222A input power/discharge is modeled internally; legacy I3 is excluded from simulation.



## Unni CO2 Smartification

Sheet: /Simulation Harness/  
File: 03\_simulation\_harness.kicad\_sch

**Title: Simulation Harness**

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| Size: A3 | Date: 2026-08-19 |
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Rev: A-v29-validated

Id: 4/4